



**College of Computing and Informatics
Computer Science Department**

Course Syllabus

Course Title

Object Oriented Design with Java

Course Code

1501 246 –Section 11

Spring Semester, 2022 / 2023

Course Syllabus

Course Title	Object Oriented Design with Java		
Course Code	1501246		
Credit Hours	3		
Pre-requisite(s)	1411211 Programming II		
Co-requisite(s)	None		
Semester	Spring	Year	2022/2023
Instructors Name	Prof Fatih Kurugollu		
Office Location	M5-217		
Tel. No.	2584		
Email	fkurugollu@sharjah.ac.ae		
Lecture Times	T R: 12:30 - 13:20 (Lectures) T : 14:00 – 15:40 (Lab)		
Office Hours	W: 14:30 – 17:30 or by appointments		

Course Description:

This course will introduce fundamental concepts of Internet technologies at both advanced and introductory levels. The introductory level includes Java programming and object-oriented (OO) design; the advanced level includes advanced programming concepts such as GUI design, multimedia, and exception handling.

Course Objectives/Goals:

The goals of this course are to enable students to:

- Understanding and using OO programming in Java.
- Understanding and implementing GUI and event handling.
- Analyzing, designing, and implementing OO-based applications.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Develop, test, and debug computer applications and applets.

2. Apply advanced concepts of OO programming such as classes and objects, array manipulation, inheritance, and polymorphism.
3. Apply the concepts of GUI programming and event handling.
4. Use and manipulate Graphics and Graphics2D objects.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	Analyze	
	Apply	- Apply advanced concepts of OO programming such as classes and objects, array manipulation, inheritance, and polymorphism. - Apply the concepts of GUI programming and event handling.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	- Develop, test, and debug computer applications and applets.
	Implement	- Develop, test, and debug computer applications and applets. - Use and manipulate Graphics and Graphics2D objects.
	Evaluate	- Develop, test, and debug computer applications and applets.
Communicate effectively in a variety of professional contexts.		
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		
Apply computer science theory and software development fundamentals to produce computing-based solutions.	Theory	- Apply advanced concepts of OO programming such as classes and objects, array manipulation, inheritance, and polymorphism. - Apply the concepts of GUI programming and event handling.
	Development	- Develop, test, and debug computer applications and applets. - Use and manipulate Graphics and Graphics2D objects.

Weekly Distribution of Course Topics/Contents

Week	Topic	Lab	Homework
1.	Introduction: Why Java?		HW#1
2.	Java Basic Elements: Variables, methods, data types, operators, input/output	Lab#1	
3.	Java Control Structures	Lab#2	
4.	Standard Class Libraries	Lab#3	HW#2
5.	User-defined Classes	Lab#4	
6.	User-defined Classes (Cont.)	Lab#5	
7.	Arrays	Lab#6	HW#3
8.	Inheritance and Polymorphism	Lab#7	
9.	ADTs and Interfaces	Lab#8	
10.	Introduction to JavaFX	Lab#9	HW#4
11.	Drawing Shapes and Text	Lab#10	
12.	Images	Lab#11	
13.	Layouts	Lab#12	
14.	Handling Events	Final Lab Exam	
15.	Revision		

Information on out-of-class assignments with due dates for submission

Homework Assignments	Due Date	Comments
HW#1	Week #4 - February 5, 2023	Covers Weeks #1,#2,#3
HW#2	Week #7 - February 26, 2023	Covers Weeks #4,#5,#6
HW#3	Week #11 - April 9, 2023	Covers Weeks #7, #8, #9
HW#4	Week #14 - April 30, 2023	Covers Weeks #10, #11, #12, #13

Students' Assessment:

Students are assessed as follows:

Assessment Tools	Date	Weight (%)	Comments
Weekly lab exercises	Every week	10	Best 12 out of 13
Programming Assignments (Homework)	Throughout the semester	10	Average of 4 homework
Midterm	Midterm Exam period	20	
Quizzes	Throughout the semester	10	Best 3 out of 4
Final Lab Exam	Week #14	10	
Final Exam	Final Exam period	40	Comprehensive
Total		100	

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Develop, test, and debug computer applications and applets.	Lectures, weekly labs, and assignments	- Lab Exercises - Quizzes - Homework - Midterm Exam - Final Lab Exam - Final Exam	Course assessment
Apply advanced concepts of OO programming such as classes and objects, array manipulation, inheritance, and polymorphism.	Lectures, weekly labs, and assignments	- Lab Exercises - Quizzes - Homework - Midterm Exam - Final Lab Exam - Final Exam	Course assessment
Apply the concepts of GUI programming and event handling.	Lectures, weekly labs, and assignments	- Lab Exercises - Quizzes - Homework - Final Lab Exam - Final Exam	Course assessment
Use and manipulate Graphics and Graphics2D objects.	Lectures, weekly labs, and assignments	- Lab Exercises - Quizzes - Homework - Final Lab Exam - Final Exam	Course assessment

Teaching and Learning Resources:

Text Book(s):

D.S. Malik. (2011). *Java Programming: From Problem Analysis to Program Design, Fifth Edition*. Course Technology, Cengage Learning. ISBN-13 978-1-111-53053-2.

Additional References:

- Deitel & Deitel. (2015). *Java: How to Program, 10th Edition*, Prentice Hall.

Software :

- **Online Compiler (GDB Online)**
<https://www.onlinegdb.com/>

This compiler can be used for basic Java programming.

- **Eclipse (Eclipse Installer 2021-12 R)**

<https://www.eclipse.org/downloads/packages/>

After launching the installation, select Eclipse IDE for Java Developers.
This compiler will be used for GUI and multimedia extensions for Java.

Laboratory Exercises:

The purpose of the laboratory exercises (supervised learning) is to help students learn Java programming in a PC environment. Starting the second week of this semester, students will have a laboratory exercise, which they should try to complete by the end of the 2-hours lab period. Before leaving the laboratory, students should submit their solutions to the Blackboard.

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive the 1st warning notice and a student missing 15% (9 hours) will receive the 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about the general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).